



BY AMETECs

DOC SR-03

Live Bug-Fix Report — Product Console (SmartEPT_Admin)

Three production errors from the 11-Aug live log — root-caused and fixed in commit `1f6be35`

Version 1.0

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1. Summary

The owner supplied the live `SmartEPT_Admin` application log for 11-Aug-2026. It contained three distinct recurring errors. Two required code fixes (both delivered in product commit `1f6be35`, 12-Aug-2026); the third was already fixed locally and simply not yet deployed. No data was lost or corrupted by any of the three.

#	Live error	Frequency	Root cause	Fix
1	<code>AttendanceDerivation</code> failed – Call to a member function <code>lessThan()</code> on string	Dozens per day, many employees	3 timestamp columns added 23-Jul never cast on the model	3 datetime casts added
2	1265 Data truncated for column <code>'action_on_blocked'</code> saving CLOSE	Each website-rules save	Two policy tables have mismatched ENUMs; shared Rules screen sends one value to both	Both columns → <code>VARCHAR(20)</code> (migration <code>2026_08_12_000200</code>)
3	Undefined array key <code>"new_employee_code"</code> on archive restore	Each restore attempt	Already fixed locally (defensive <code>?? null</code>) — commit not yet on live	Deploys with this push; no new code

2. Bug 1 — Attendance derivation crashing (missing casts)

What was happening. Every attendance recalculation for an employee with agent activity failed. The QA-Phase-2 migration (23-Jul) added `first_login_at`, `last_login_at` and `final_logout_at` to `employee_attendance_logs`, but `EmployeeAttendanceLog::$casts` was never updated — the values arrived as plain strings, and the first Carbon comparison (`lessThan`) inside `AttendanceDerivation::earliest()/latest()` threw. Attendance rows for those employees silently stopped updating.

Fix. The three missing `datetime` casts on the model. Derivation is re-entrant, so every affected employee **self-heals on the next run** after deployment — no repair script needed.

Team lesson (recorded in project memory). Adding a timestamp column and adding its model cast are ONE change — never separate commits.

3. Bug 2 — Policy save 1265 (ENUM strike four)

What was happening. `application_policies.action_on_blocked` allows `CLOSE` (not `BLOCK`); `website_policies.action_on_blocked` allows `BLOCK` (not `CLOSE`). The console's shared Rules screen sends the SAME chosen action to BOTH tables — so choosing `CLOSE` saved the application policy and crashed the website policy with strict-MySQL 1265 (and `BLOCK` would fail the opposite way). This is the **fourth** live path broken by an ENUM.

Fix. Migration `2026_08_12_000200` converts both columns to `VARCHAR(20) DEFAULT 'WARN'` per the standing prefer-VARCHAR-over-ENUM rule (sqlite-guarded for the test suite; rollback restores a union ENUM so no row is ever truncated).

4. Bug 3 — Archive restore (already fixed, undeployed)

The live error at `EmployeeController::restoreArchive` matches code that is **already fixed in the local tree** — a defensive `$data['new_employee_code'] ?? null` with a comment describing precisely this failure. Live simply has not received that commit yet. It reaches the VPS with this same push; nothing new was written.

5. Deployment & Verification

Local: product `migrate.bat` → Laragon Stop All/Start All → `git push`. **Live:** `git pull && php artisan migrate --force && php artisan config:clear` → reload php8.3-fpm. (One `migrate --force` also applies the pending mail-settings/OTP migration — two migrations, one command.)

Verify after deploy: (a) `laravel.log` stays free of `AttendanceDerivation failed` warnings and today's attendance rows update for the previously affected employees (10, 11, 15, 18, 32, 36, 37, 42); (b) Rules screen saves with action `CLOSE` **and** `BLOCK` without error, policy version increments; (c) an archived employee restores cleanly with no new code typed. These three checks have been added to the team's test scope alongside the BF suite.